

**Lab sheet 20****Submission Due: 04.11.2024****Title:** E – Commerce App (React) - I**Aims:**

- Implement a ReactJS frontend application

**Tasks:**

- Create a New Project
- Install Necessary Packages
- Implement Components
- Integrate Components into the Page

**Activities :**

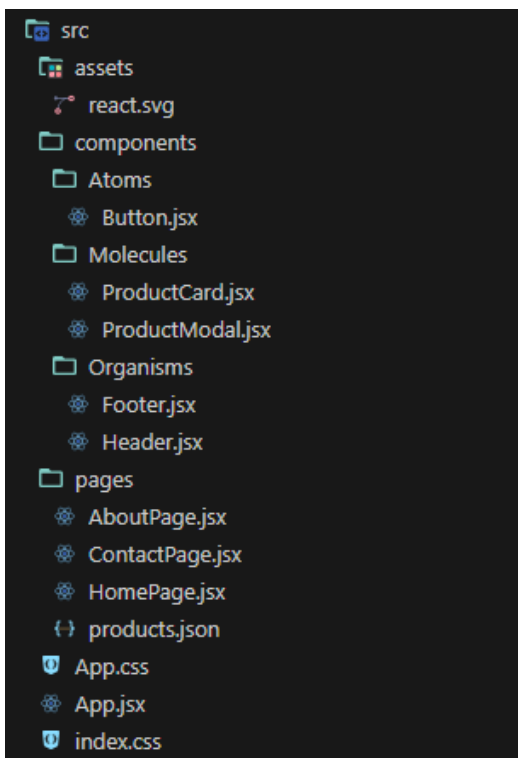
1. Create new project named “e-commerce-app”
2. Install Bootstrap

***npm install bootstrap******npm install react-bootstrap bootstrap***

3. Install react-route-dom

***npm install react-router-dom***

4. Create folder structure as follows



## 5. Implement Footer.jsx

```
import { Navbar, Container } from "react-bootstrap";

const Footer = () => {
  return (
    <Navbar bg="dark" variant="dark" fixed="bottom">
      <Container>
        <p className="text-white">Copyright &copy; 2024 My E-commerce
App</p>
      </Container>
    </Navbar>
  );
};

export default Footer;
```

## 6. Implement Header.jsx

```
import { Navbar, Container, Nav } from 'react-bootstrap';
import { Link } from 'react-router-dom';

const Header = () => {
  return (
    <Navbar bg="dark" variant="dark" fixed='top'>
      <Container>
        <Navbar.Brand href="#home">My E-commerce App -
Admin</Navbar.Brand>
        <Nav>
          <Nav.Link as={Link} to="/">Products</Nav.Link>
          <Nav.Link as={Link} to="/about">About</Nav.Link>
          <Nav.Link as={Link} to="/contact">Contact</Nav.Link>
        </Nav>
      </Container>
    </Navbar>
  );
};

export default Header;
```

## 7. Implement ProductCard.jsx

```
import Card from 'react-bootstrap/Card';
const ProductCard = ({ product }) => {
  return (
    <Card>
      <Card.Img variant="top" src={product.image} />
      <Card.Body>
        <Card.Title>{product.name}</Card.Title>
        <Card.Text>{product.description}</Card.Text>
      </Card.Body>
    </Card>
  );
};
```

```

        <Card.Text>${product.price}</Card.Text>
      </Card.Body>
    </Card>
  );
};

export default ProductCard;

```

#### 8. Implement ProductModal.jsx

```

import { useState } from "react";
import { Button, Modal, Form } from "react-bootstrap";

const ProductModal = ({ show, onHide, onSubmit, initialProduct }) => {
  const [formData, setFormData] = useState(
    initialProduct || {
      name: "",
      description: "",
      price: "",
    }
  );

  const handleChange = (e) => {
    setFormData({ ...formData, [e.target.name]: e.target.value });
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    onSubmit(formData);
    onHide();
  };

  return (
    <Modal show={show} onHide={onHide}>
      <Modal.Header closeButton>
        <Modal.Title>
          {initialProduct ? "Edit Product" : "Add Product"}
        </Modal.Title>
      </Modal.Header>
      <Modal.Body>
        <Form onSubmit={handleSubmit}>
          <Form.Group controlId="formProductName">
            <Form.Label>Product Name</Form.Label>
            <Form.Control
              type="text"
              placeholder="Enter product name"
              value={formData.name}
              onChange={handleChange}
              required

```

```

        />
      </Form.Group>
    </Form>
  </Modal.Body>
  <Modal.Footer>
    <Button variant="secondary" onClick={onHide}>
      Close
    </Button>
    <Button variant="primary" type="submit">
      {initialProduct ? "Save Changes" : "Add Product"}
    </Button>
  </Modal.Footer>
</Modal>
);
};

export default ProductModal;

```

### 9. Implement AboutPage.jsx

```

import Header from '../components/Organisms/Header';
import Footer from '../components/Organisms/Footer';

const AboutPage = () => {
  return (
    <>
      <Header />
      <div className="mx-5">
        <h1>About Us</h1>
        <p>
          Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed
          do eiusmod tempor incididunt ut labore et dolore magna aliqua.
          Ut enim ad
          minim veniam, quis nostrud exercitation ullamco laboris nisi
          ut
          aliquip ex ea commodo consequat. Duis aute irure dolor in
          reprehenderit in voluptate velit esse cillum dolore eu
          fugiat nulla
          pariatur. Excepteur sint occaecat cupidatat non proident,
          sunt in
          culpa qui officia deserunt mollit anim id est laborum.
        </p>
        <p>
          <strong>Our Mission:</strong> To provide high-quality
          products and
          exceptional customer service.
        </p>
      </div>
    </>
  );
};

```

```

        <strong>Our Vision:</strong> To be the leading e-commerce
platform in
        the industry.
    </p>
</div>
<Footer />
</>
);
};

export default AboutPage;

```

10. Implement ContactPage.jsx

11. Implement HomePage.jsx

```

import Header from "../components/Organisms/Header";
import Footer from "../components/Organisms/Footer";
import ProductCard from "../components/Molecules/ProductCard";
import productsData from "../products.json";
import { useState } from "react";
import { Button } from "react-bootstrap";
import ProductModal from "../components/Molecules/ProductModal";

const HomePage = () => {
    const [showProductModal, setShowProductModal] = useState(false);
    const [currentProduct, setCurrentProduct] = useState(null);

    const handleAddProduct = () => {
        setCurrentProduct(null);
        setShowProductModal(true);
    };

    const handleEditProduct = (product) => {
        setCurrentProduct(product);
        setShowProductModal(true);
    };

    const handleProductSubmit = (productData) => {
        setShowProductModal(false);
    };

    return (
        <>
            <Header />
            <div className="mt-5">
                <Button onClick={handleAddProduct}>Add Product</Button>
                <ProductModal
                    show={showProductModal}
                    onHide={() => setShowProductModal(false)}

```

```

        onSubmit={handleProductSubmit}
        initialProduct={currentProduct}
      />
      <h1 className="text-center mt-4">All Products</h1>
      <div className="row mb-3">
        {productsData.map((product) => (
          <div className="col-md-3 mb-3" key={product.id}>
            <ProductCard product={product} />
            <Button onClick={() => handleEditProduct(product)}
className="mt-2">
              Edit Product
            </Button>
          </div>
        ))}
      </div>
    </div>
    <Footer />
  </>
);
};

export default HomePage;

```

## 12. Implement App.jsx

```

import { BrowserRouter as Router, Routes, Route } from 'react-router-
dom';
import HomePage from './pages/HomePage';
import AboutPage from './pages/AboutPage';
import ContactPage from './pages/ContactPage';
import './App.css';

const App = () => {
  return (
    <Router>
      <Routes>
        <Route path="/" element={<HomePage />} />
        <Route path="/about" element={<AboutPage />} />
        <Route path="/contact" element={<ContactPage />} />
      </Routes>
    </Router>
  );
};

export default App;

```